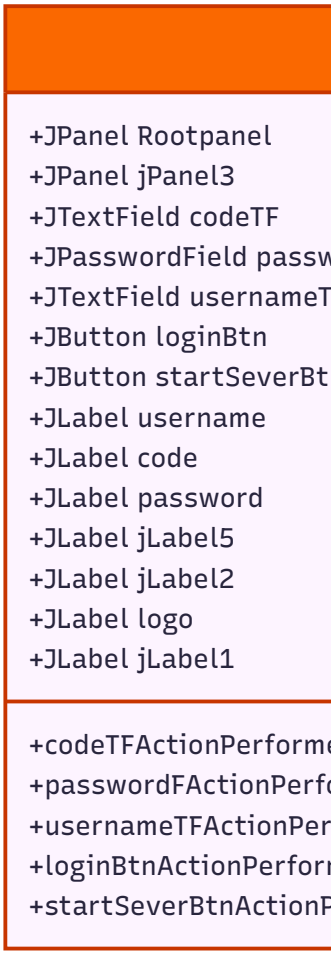
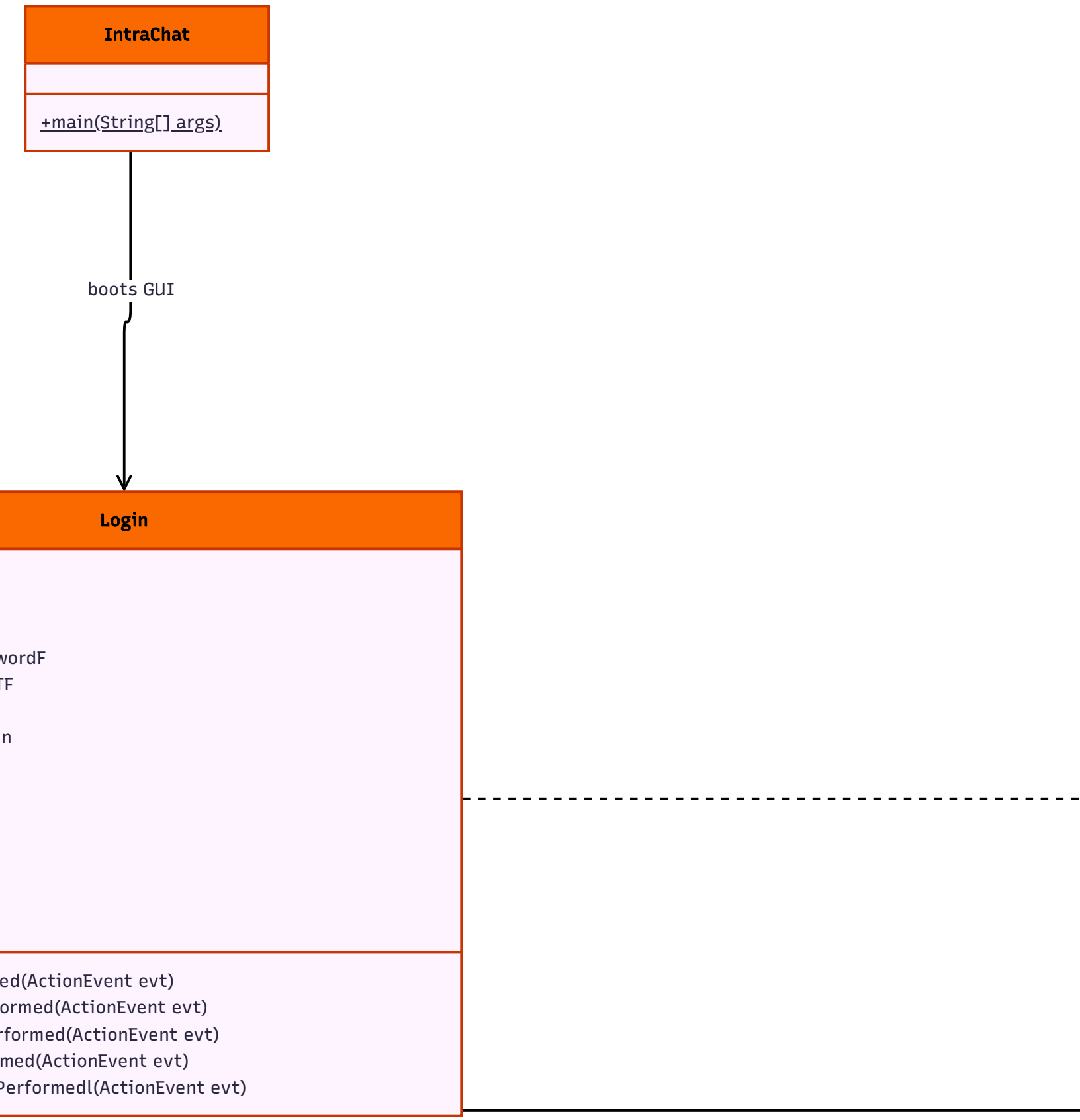
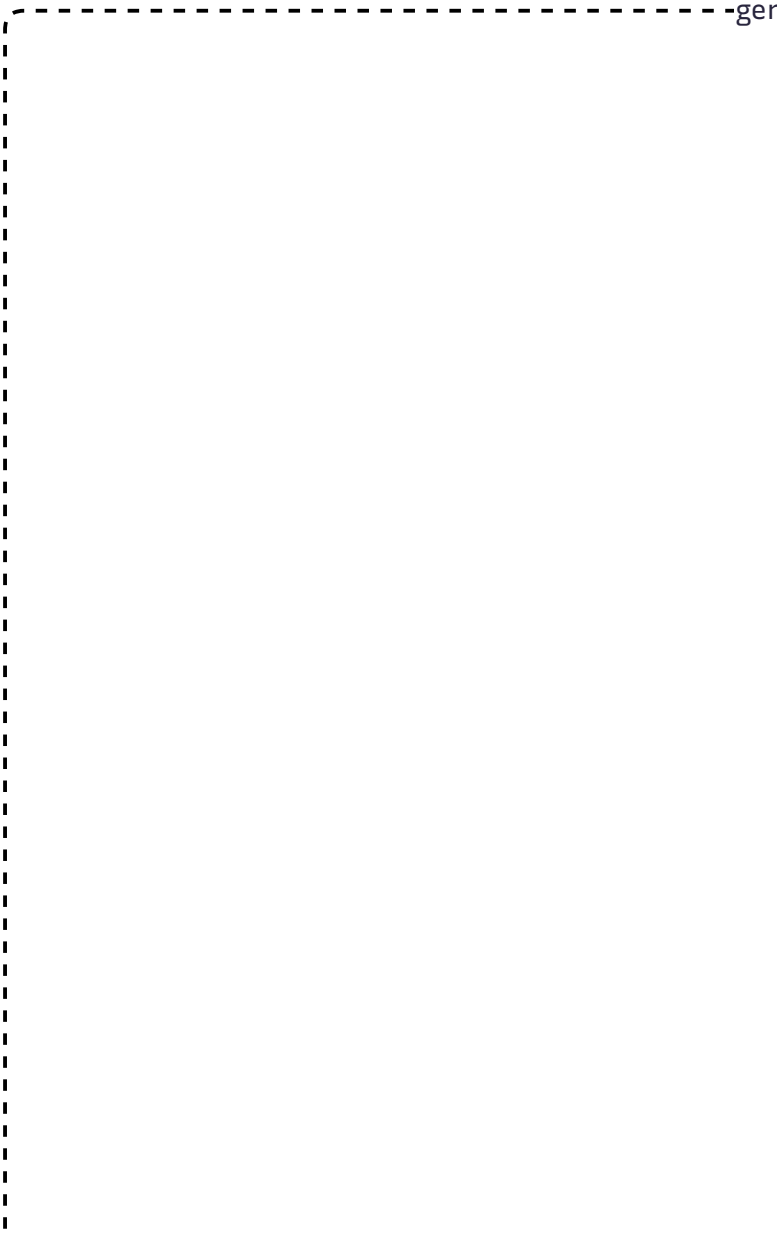
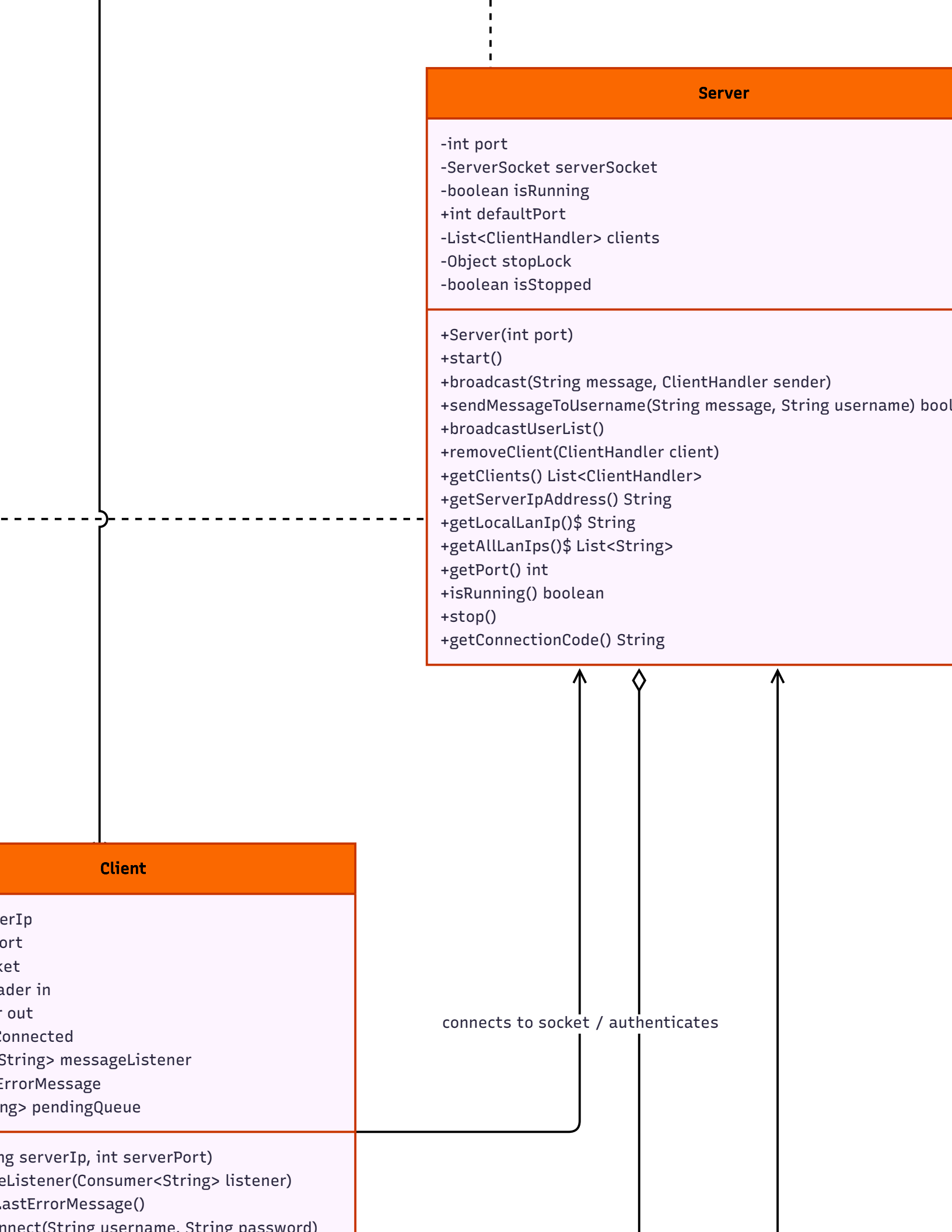

+JPanel Rootpanel
+JPanel jPanel3
+JTextField codeTF
+JPasswordField passw
+JTextField usernameT
+JButton loginBtn
+JButton startSeverBt
+JLabel username
+JLabel code
+JLabel password
+JLabel jLabel5
+JLabel jLabel2
+JLabel logo
+JLabel jLabel1
+codeTFActionPerforme
+passwordFActionPerfo
+usernameTFActionPer
+loginBtnActionPerfor
+startSeverBtnActionP

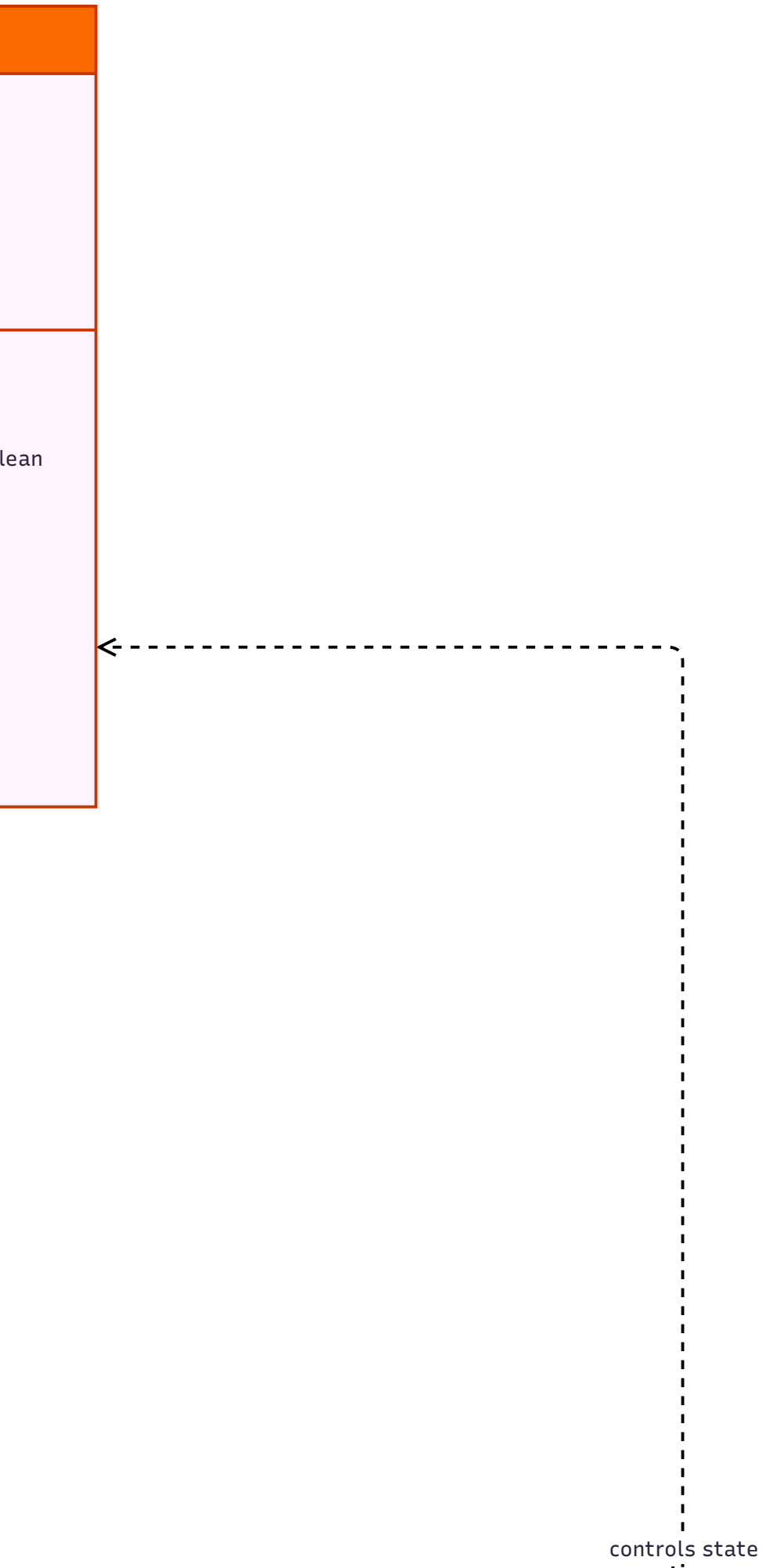

IntraChat
+main(String[] args)
boots GUI
Login
wordF
TF
n
ed(ActionEvent evt)
ormed(ActionEvent evt)
rformed(ActionEvent evt)
med(ActionEvent evt)
Performedl(ActionEvent evt)

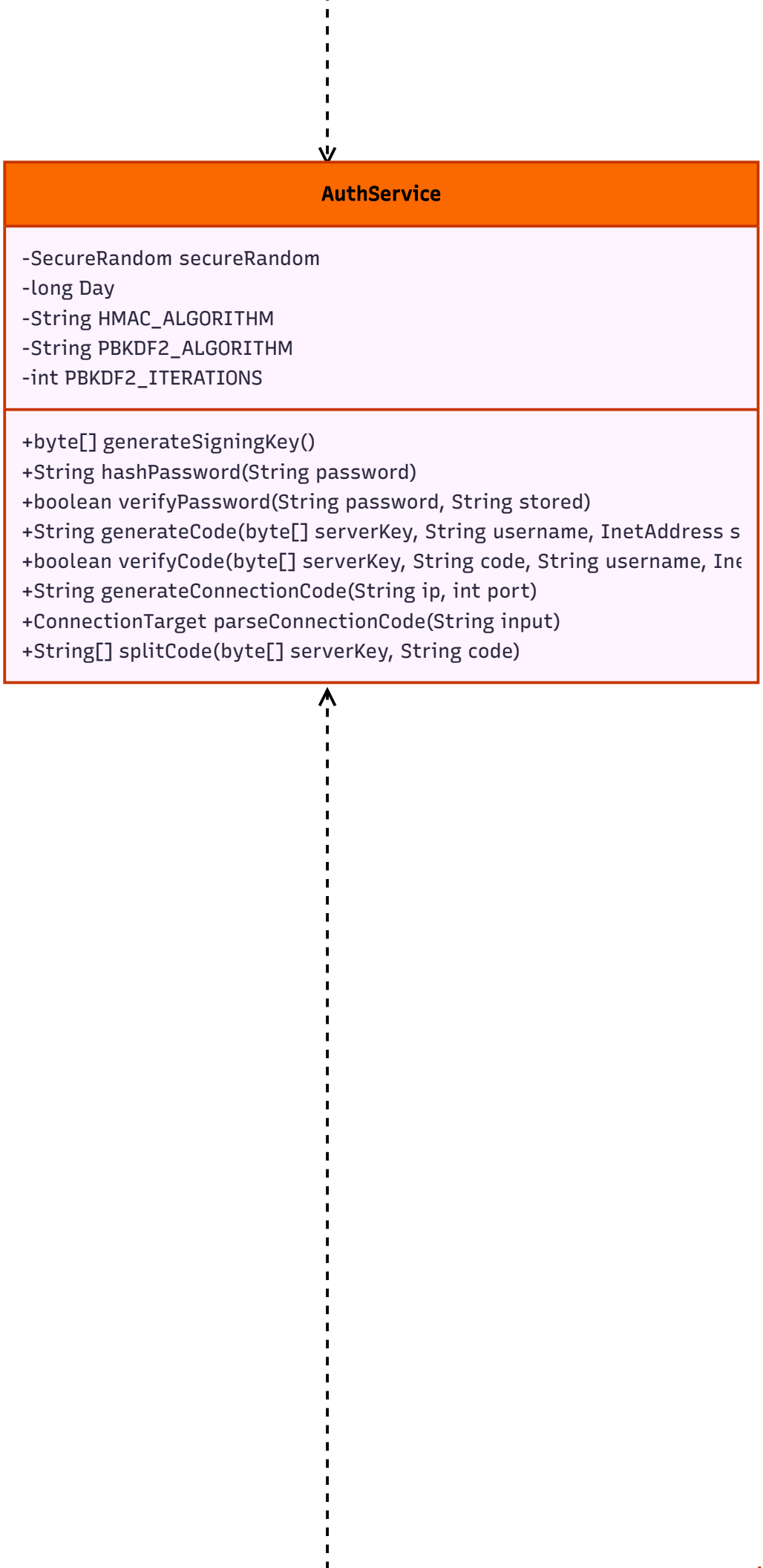

authenticates via socket
initializes connection

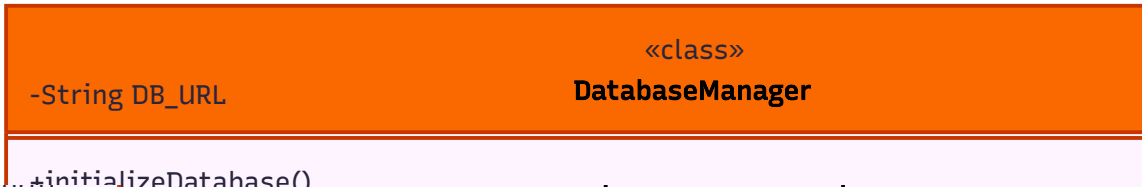


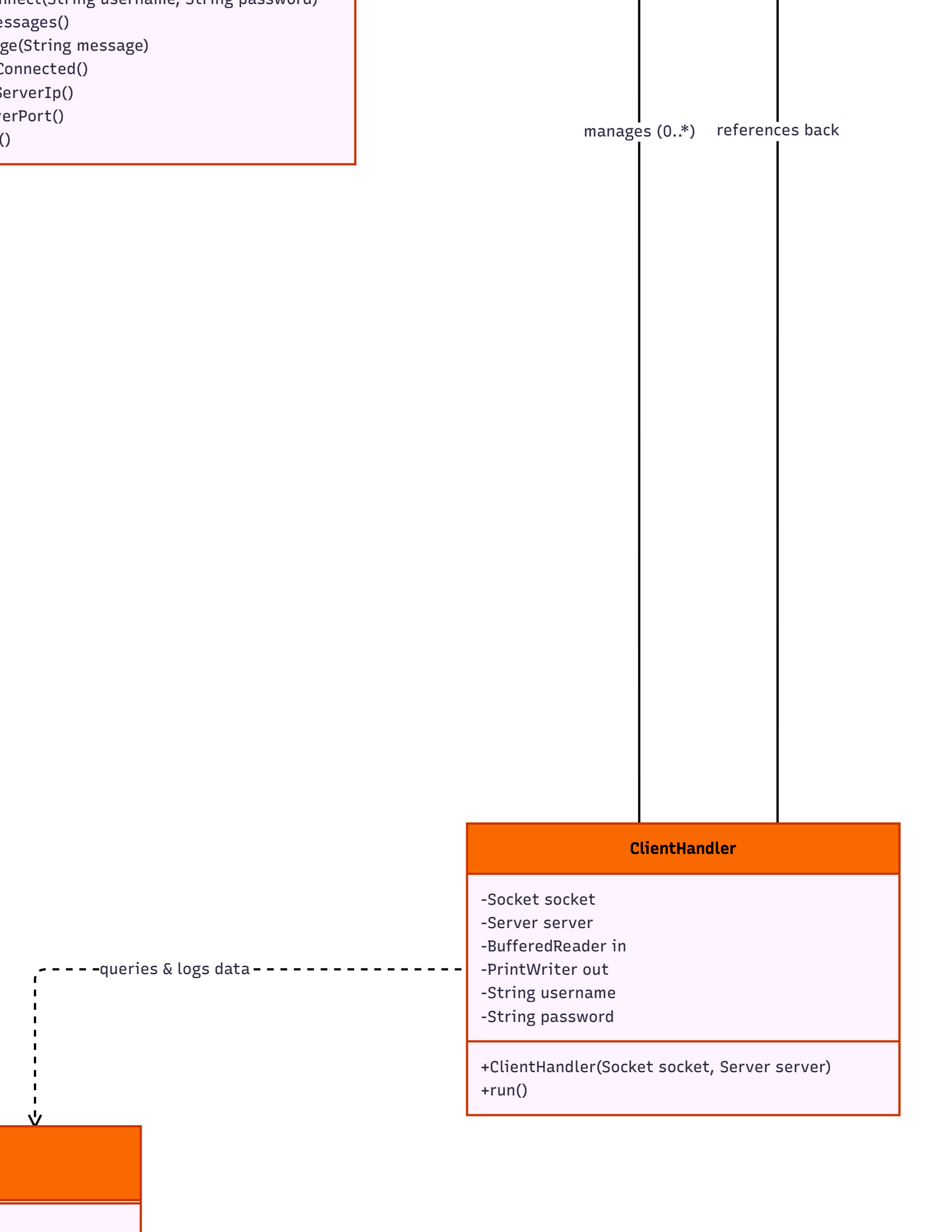

nerates/verifies codes
-String serve
-int serverPo
-Socket sock
-BufferedRea
-PrintWriter
-boolean isC
-Consumer<S
-String lastE
-Queue<Strin
+Client(Strin
+setMessage
+String getL
+boolean co

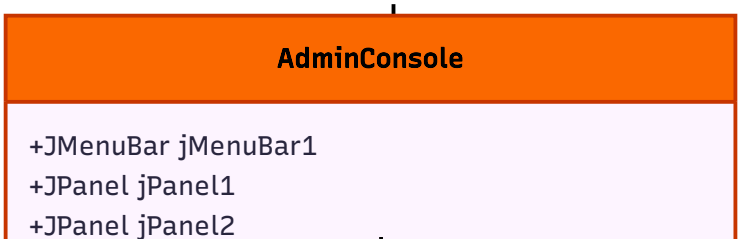

Server
-int port
-ServerSocket serverSocket
-boolean isRunning
+int defaultPort
-List<ClientHandler> clients
-Object stopLock
-boolean isStopped
+Server(int port)
+start()
+broadcast(String message, ClientHandler sender)
+sendMessageToUsername(String message, String username) bool
+broadcastUserList()
+removeClient(ClientHandler client)
+getClients() List<ClientHandler>
+getServerIpAddress() String
+getLocalLanIp()$ String
+getAllLanIps()$ List<String>
+getPort() int
+isRunning() boolean
+stop()
+getConnectionCode() String
Client
erIp
ort
ket
ader in
out
Connected
String> messageListener
ErrorMessage
ng> pendingQueue
ng serverIp, int serverPort)
eListener(Consumer<String> listener)
LastErrorMessage()
nnect(String username, String password)
connects to socket / authenticates

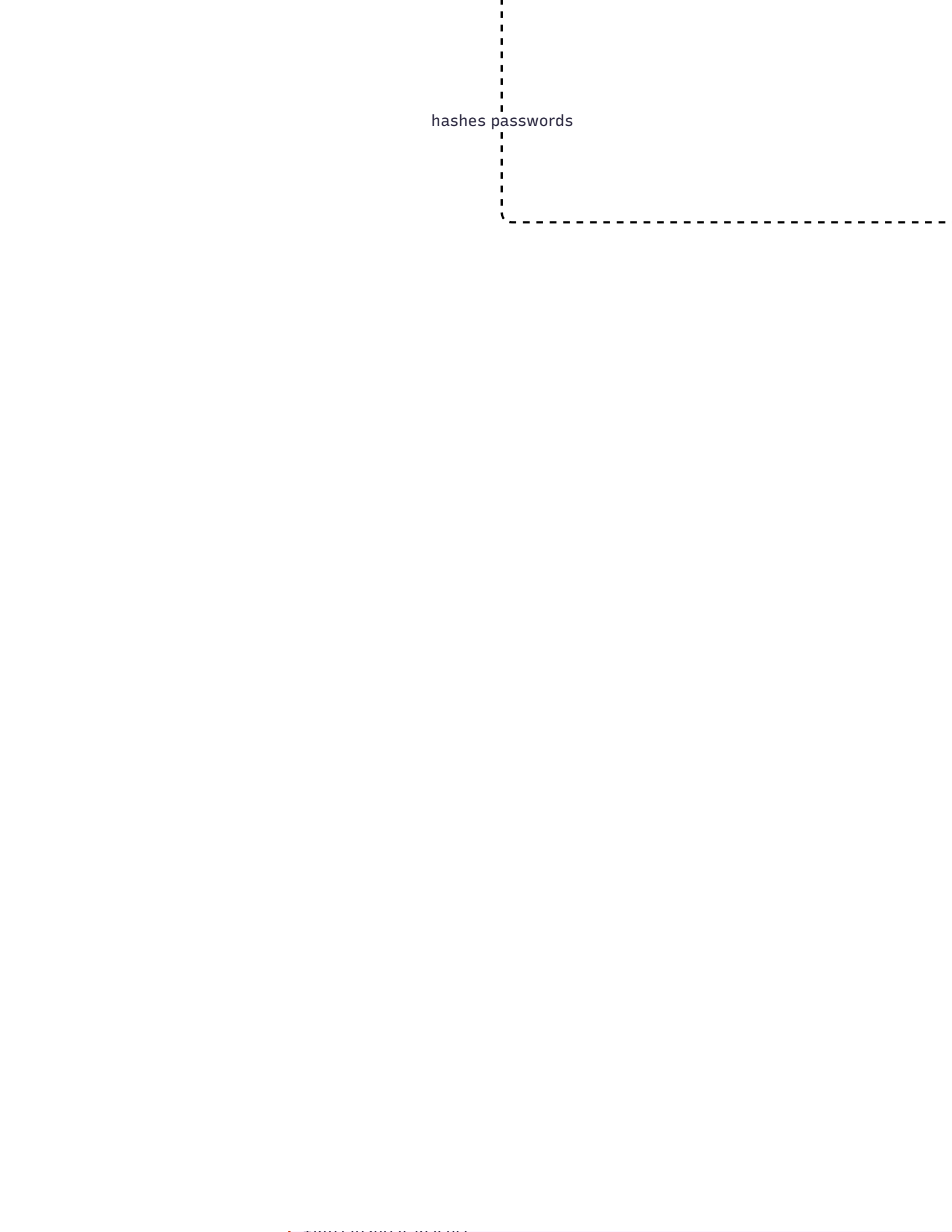

lean
controls state

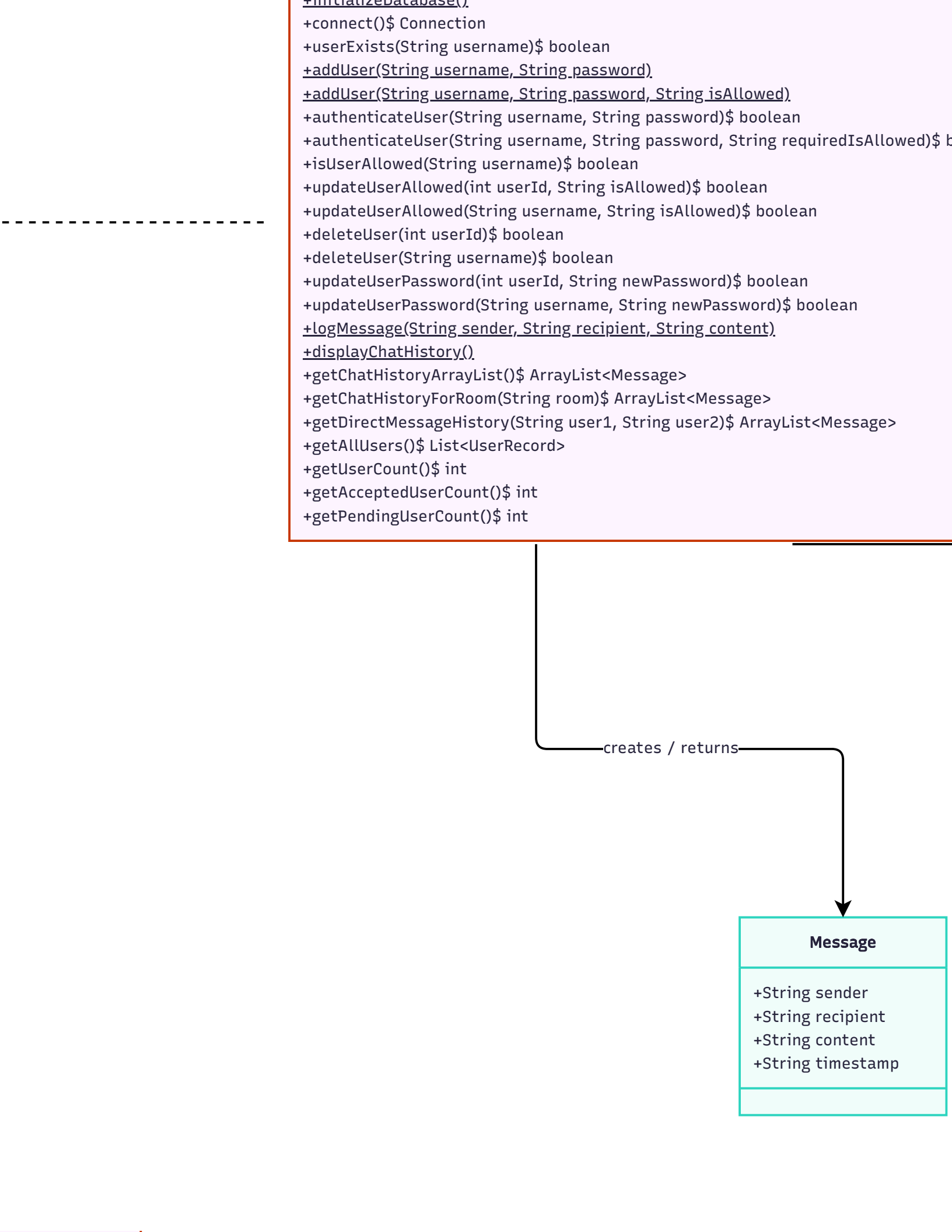

AuthService
-SecureRandom secureRandom
-long Day
-String HMAC_ALGORITHM
-String PBKDF2_ALGORITHM
-int PBKDF2_ITERATIONS
+byte[] generateSigningKey()
+String hashPassword(String password)
+boolean verifyPassword(String password, String stored)
+String generateCode(byte[] serverKey, String username, InetAddress s
+boolean verifyCode(byte[] serverKey, String code, String username, Ine
+String generateConnectionCode(String ip, int port)
+ConnectionTarget parseConnectionCode(String input)
+String[] splitCode(byte[] serverKey, String code)

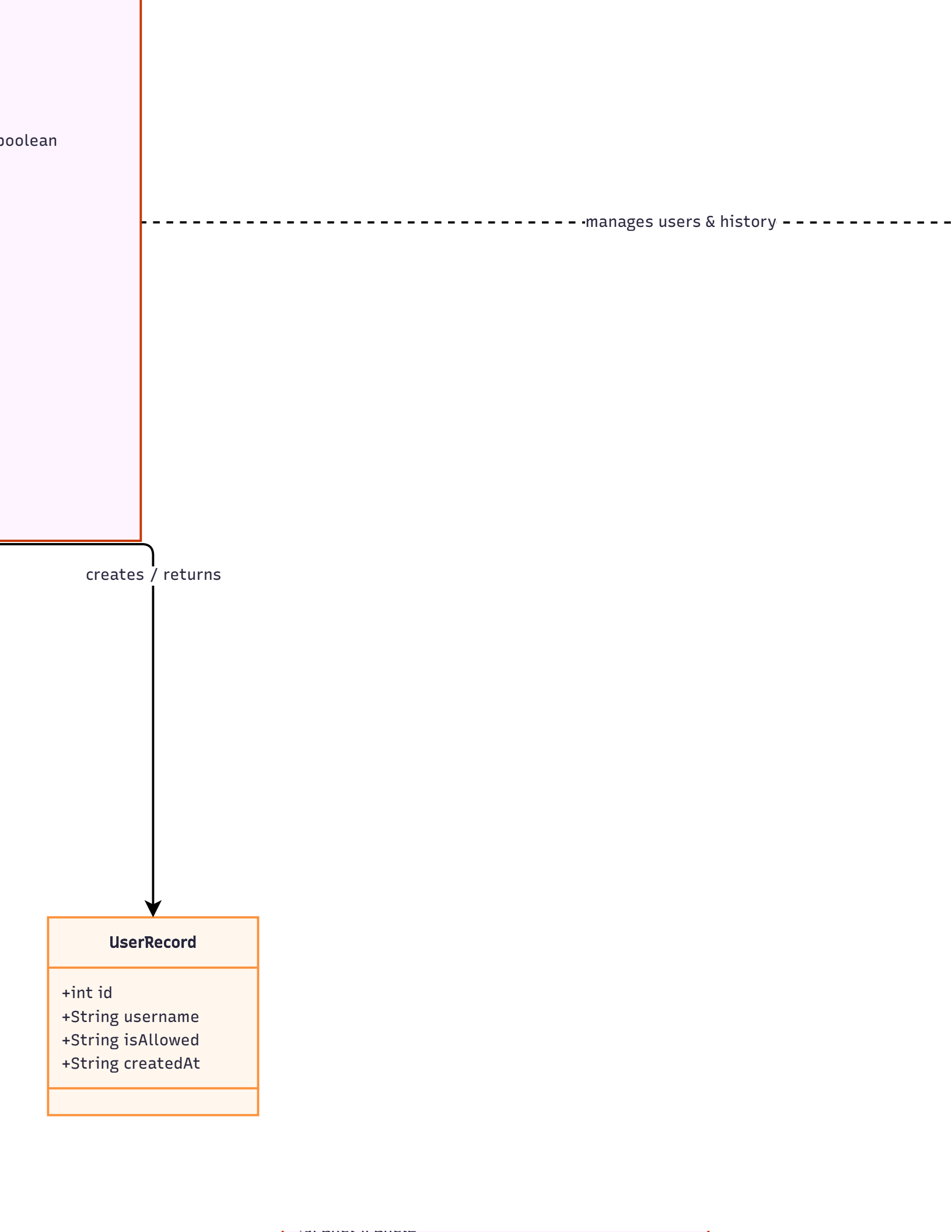

-listenForMe
+sendMessag
+boolean isC
+String getS
+int getServ
+disconnect(
«class»
DatabaseManager
-String DB_URL
+initializeDatabase()

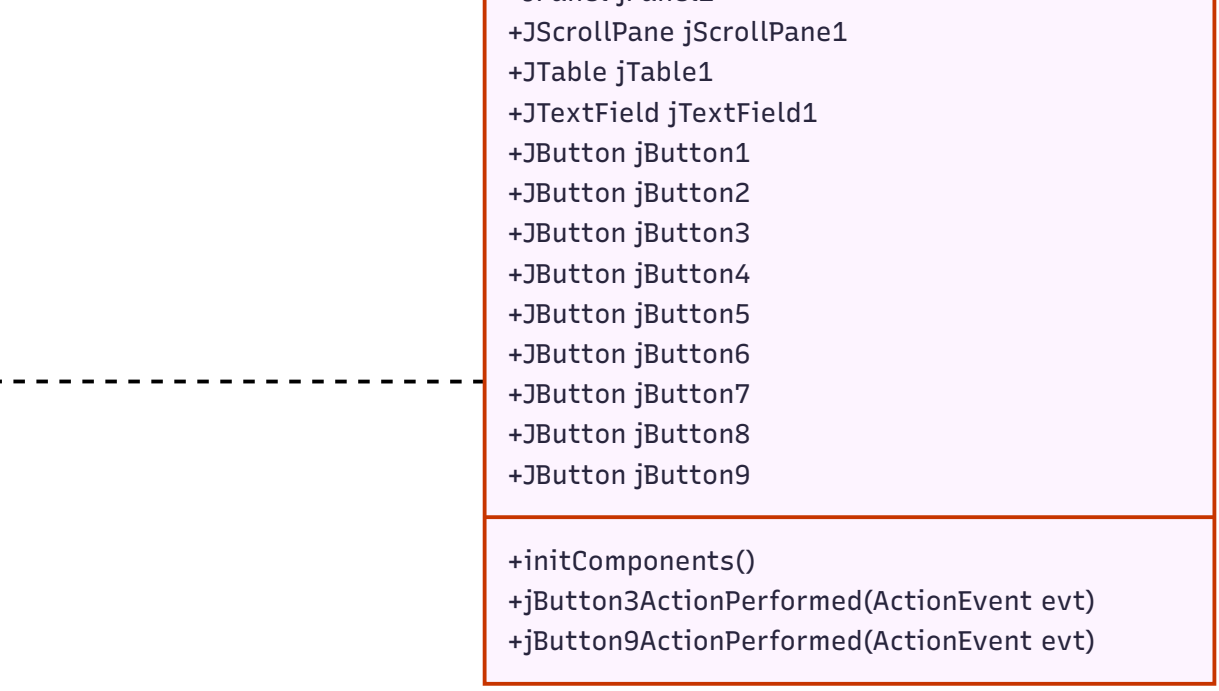

essages()
ge(String message)
Connected()
ServerIp()
erPort()
()
manages (0..*)
references back
ClientHandler
-Socket socket
-Server server
-BufferedReader in
-PrintWriter out
-String username
-String password
+ClientHandler(Socket socket, Server server)
+run()
queries & logs data


AdminConsole
+JMenuBar jMenuBar1
+JPanel jPanel1
+JPanel jPanel2


hashes passwords


+initializeDatabase()
+connect()$ Connection
+userExists(String username)$ boolean
+addUser(String username, String password)
+addUser(String username, String password, String isAllowed)
+authenticateUser(String username, String password)$ boolean
+authenticateUser(String username, String password, String requiredIsAllowed)$ b
+isUserAllowed(String username)$ boolean
+updateUserAllowed(int userId, String isAllowed)$ boolean
+updateUserAllowed(String username, String isAllowed)$ boolean
+deleteUser(int userId)$ boolean
+deleteUser(String username)$ boolean
+updateUserPassword(int userId, String newPassword)$ boolean
+updateUserPassword(String username, String newPassword)$ boolean
+logMessage(String sender, String recipient, String content)
+displayChatHistory()
+getChatHistoryArrayList()$ ArrayList<Message>
+getChatHistoryForRoom(String room)$ ArrayList<Message>
+getDirectMessageHistory(String user1, String user2)$ ArrayList<Message>
+getAllUsers()$ List<UserRecord>
+getUserCount()$ int
+getAcceptedUserCount()$ int
+getPendingUserCount()$ int
creates / returns
Message
+String sender
+String recipient
+String content
+String timestamp


boolean
manages users & history
creates / returns
UserRecord
+int id
+String username
+String isAllowed
+String createdAt


+JScrollPane jScrollPane1
+JTable jTable1
+JTextField jTextField1
+JButton jButton1
+JButton jButton2
+JButton jButton3
+JButton jButton4
+JButton jButton5
+JButton jButton6
+JButton jButton7
+JButton jButton8
+JButton jButton9
+initComponents()
+jButton3ActionPerformed(ActionEvent evt)
+jButton9ActionPerformed(ActionEvent evt)